Sandeep Rajansingh Reswal

mp24mscst14007@iith.ac.in | reswal.sandeep@gmail.com | +91 9511737404 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.64
Fr. Agnel Multipurpose Higher Secondary School, Verna - Goa	Class XII (Goa Board Of Secondary and Higher Secondary Education)	2014	73.00%
Our Lady Of Rosary High School Fatorda, Salcete - Goa	Class X (Goa Board Of Secondary and Higher Secondary Education)	2012	73.17%

SCHOLASTIC ACHIEVEMENTS

Qualified **JAM** *(Joint Admission Test for Masters)* in **Physics**, a National-Level Examination, in **2024, 2023,** and **2021**

Qualified among the **Top 1.5%** of applicants in the UPSC **CDSE** written examination across **seven attempts** (2017-2020)

Qualified among the **Top 2.4%** of 259,640 candidates in the UPSC **NDA & NA (II)**, **2014** National Level Examination

Cleared **Air Force Common Admission Test** written examination in **2021, 2019,** and **2018,** among 150,000-200,000 applicants per cycle

PROJECTS

Project-D: Cyclotron for Education & Research

- Identified a **gap in Indian accelerator infrastructure** and designed a **compact 0.5 MeV proton cyclotron**; secured **INR 0.5 lakh** through **IIT Hyderabad's BUILD Program**, with the project selected among **40 funded proposals** from **100+ applications**.
- **CAD-modeled** the complete assembly in **Solid Edge** and simulated core physics using **Poisson-Superfish, Opera-3D, and MATLAB**; built and **validated a Python-based particle-orbit simulator** using **NumPy/SciPy** for computational design validation.
- Exploring **Physics-Informed Neural Networks (PINNs)** for magnetostatic field prediction, alongside **scikit-learn surrogate modeling and Bayesian optimization** for automated design-parameter search; extending the workflow to **COMSOL**-based multiphysics simulation on CAD geometry.

PelviContour - AI-Assisted Organ-at-Risk Contouring for Radiotherapy Planning [Streamlit] [github]

- Built and trained a **3D U-Net convolutional neural network** using **PyTorch/MONAI** to automatically segment bladder and bowel from pelvic CT scans for OAR delineation in gynecological brachytherapy; achieved **0.68 Dice for bladder and 0.55 for bowel** on held-out test cases.
- **Validated TotalSegmentator as a training proxy** in the absence of public brachytherapy-specific segmentation datasets, and **characterized the domain gap** against published clinical benchmarks.
- **Deployed an interactive Streamlit application** to visualize predicted contours and estimate department-level time and cost impact, translating model performance into **potential radiotherapy workflow capacity gains**.

Real Estate Price Predictor & Recommender [streamlit] [github]

- Built an **end-to-end ML pipeline** for Gurgaon real estate, transforming **26K+ scraped listings into 3,554 modeling-ready properties** through **data cleaning, feature engineering,** and **EDA**
- **Benchmarked 8 feature-selection** and **4 encoding techniques** across Linear Regression, SVR, and Random Forest; optimized the final Random Forest using RandomizedSearchCV, achieving **0.91 test R-squared** and **INR 17-23 lakh MAE** for sub-INR 2 Cr properties, with **SHAP-based model explainability**
- Developed a **multi-page Streamlit application** integrating price prediction and a **cosine-similarity recommendation engine**, landmark-proximity search, and interactive market analytics across **246 residential projects and 1,070 landmarks**

INTERNSHIPS

- *Basavatarakam Indo American Cancer Hospital & Research Institute - Hyderabad:* **Medical Physics Intern** - Clinical QA, treatment planning, and delivery across CT Simulation, TPS, HDR Brachytherapy, and TomoTherapy
- *SAMEER (Autonomous R&D Lab under MeitY, Govt of India)* - Studied LINAC-based radioisotope production and cyclotron magnet optimization using **Opera-3D, Poisson-Superfish, and MATLAB** under SAMEER's Medical Systems Division

SKILLS

- **Languages & Tools**: Python, MATLAB, Git, GitHub Actions, Jupyter Notebook, VS Code, SQL, LaTeX
- **Libraries/Frameworks**: Pandas, NumPy, Matplotlib, SciPy, Scikit-learn, SHAP, PyTorch, MONAI, Streamlit, Plotly, Seaborn
- **Core Competencies**: Machine Learning, Data Cleaning & EDA, Data Visualization, ETL Pipelines, Model Explainability, Medical Physics (Radiation Dosimetry, Treatment Planning, QA)
- **Soft Skills**: Problem Solving, Cross-Domain Collaboration, Technical Documentation, Data-Driven Decision Making

RELEVANT COURSES

- Quantum Mechanics, Mathematical Physics, Differential Equations, Numerical Methods, Computational Techniques, Electrodynamics & Electromagnetic Theory, Solid State Physics, Thermodynamics & Statistical Mechanics, Atomic and Molecular Physics, Computational Modeling of Biological Systems, Basic Algebra, Computers for Math, Statistical Methods, Operations Research, Radiological Mathematics, Medical Imaging

POSITIONS OF RESPONSIBILITY

- Placement Coordinator, **Office of Career Services, IIT Hyderabad -** 2025 - Present

- Coach, **Tiger's Taekwondo Club** - **Mentored team** to **Overall Championship victory**, Goa University Inter-College Championship.

- **University Faculty Representative** - Represented the **Faculty of Science** at university-level activities and forums

AWARDS

- Finalist, *LaunchPad Challenge* **- Business Model Competition**, South India Edition 2024, XIME Bangalore
- Received the **Departmental Certificate of Merit** for being the **Top-Performing First-Year Student** in **Physics** - 2016
- Samraat Student of the Year 2010-11, **State Award of Excellence**, Samraat Students Foundation, **Goa** - Awarded for being the **Most Cultured, Disciplined, and Best Student** at Our Lady of Rosary High School, recognized as an *ideal citizen of tomorrow*

SPORTS

Taekwondo **Black Belt** with multiple Gold, Silver & Bronze medals at **State, West Zone, Inter-Zone** and **National-level** championships; represented **Goa at National School Games & National Championships;** IIT Hyderabad **Half Marathon Finisher - 21.1 km**

HOBBIES AND INTERESTS

- I find comfort in reading novels, watching documentaries, and movies, discovering new ideas and perspectives that make me see the world a little differently